

2022-Ph-H1-Q11

Section: Our Dynamic Universe

Topic: Energy and Power

A car accelerates and gains kinetic energy. The power output of the engine is calculated. What is the definition of average power?

Answer: D

(Average power = energy ÷ time)

Guidance for Students:

Power is defined as the **rate of energy transfer**. This question uses the average case — total energy gained over time.

Revision Tips:

- Average power: $P = \frac{E}{t}$
- Units: Watts (W) = Joules per second (J/s)
- Can also use $P = Fv$ if force and velocity are constant